PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

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Abstract

Flooding is one of the most frequent and damaging natural hazards globally, and its risk is projected to increase due to climate change. Accurate flood risk assessments are essential for spatial planning, climate adaptation, and ESG-aligned financial decision-making. This thesis investigates physical flood risk by comparing global and local hazard models, evaluating standardized vulnerability functions, and quantifying financial implications for real estate assets in Slovenia.

Two flood hazard models are assessed: the global WRI Aqueduct River Flood model and the high-resolution local model Integralna karta globin (IKG). A geospatial validation dataset comprising flood depth and damage records from four flood events in Slovenia (2010–2023) is used to evaluate model accuracy. Flood depths are analyzed using both regression and classification metrics, while vulnerability is assessed through comparison of observed damages against standardized depth–damage functions provided by the European Commission Joint Research Centre (JRC).

A key technical contribution is the development of a data processing pipeline that harmonizes heterogeneous IKG hazard data and integrates it into the modular Physrisk platform using the Zarr data format. This enables scalable, scenario-based risk assessment and model intercomparison.

Findings suggest that while local models offer improved accuracy for continuous flood depth prediction in regression tasks, the global model performs better in categorical classification. Proximity to water bodies was not found to significantly improve prediction accuracy. Depth–damage functions align well with observed damage distributions for residential properties, though segmentation by property type reveals variability in model performance.

The thesis contributes to the validation and integration of physical climate risk models for real estate applications and enables more accurate, location-specific flood risk estimates to support planning and investment decisions.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

Contents

List of Figures xiii

List of Tables xv

List of Algorithms xvii

Abbreviations xix

Symbols xxi

Glossary xxiii

1 Introduction 1
- 1.1 Background and Motivation . . . 1
- 1.2 Climate Risk and ESG Disclosure . . . 1
- 1.3 Problem Statement . . . 2
- 1.4 Research Objectives . . . 2
- 1.5 Structure of the Thesis . . . 3

2 Literature Review & Related Work 5

3 Study Area, Data, and Methodology 7
- 3.1 Hazard and Depth . . . 7
  - 3.1.1 Mathematical Definition of Flood Hazard . . . 7
  - 3.1.2 Hazard Data Sources and Modeling Approach . . . 8
- 3.2 Vulnerability and Damage Function . . . 9
  - 3.2.1 Mathematical Definition of Flood Vulnerability . . . 9
  - 3.2.2 Vulnerability Data Source . . . 10
- 3.3 Flood Risk Modeling . . . 10
  - 3.3.1 Risk for a Single Return Period . . . 10
  - 3.3.2 Expected Annual Damage (EAD) . . . 11
  - 3.3.3 Exposure Data . . . 11
  - 3.3.4 Use Cases . . . 12

4 Model Integration and Analysis Framework 13
- 4.1 Architectural Diagram . . . 13
  - 4.1.1 System Overview . . . 14
  - 4.1.2 Data Flow and Model Interoperability . . . 14
  - 4.1.3 Integration of the IKG Model . . . 14
  - 4.1.4 Hazard Onboarding Pipeline . . . 14
- 4.2 Pipeline for Combining Heterogeneous IKG Models . . . 15
  - 4.2.1 Data Sources and Heterogeneity . . . 15

4.2.2 Standardization and Processing Workflow 15
4.2.3 Unified Flood Hazard Product 16
4.3 Physrisk Framework – Model Onboarding 16
4.3.1 Physrisk Architecture and Modules 16
4.3.2 Zarr Format and Indexing 17
4.3.3 Onboarding Procedure 17
4.3.4 Integration with Other Models 17
4.4 Comparison Toolset 18
4.4.1 Purpose and Design Principles 18
4.4.2 Interface and Functionality 18
4.4.3 Supported Models and Inputs 18
4.4.4 Integration with Validation Workflow 19
4.4.5 Reproducibility and Extensibility 19

5 Experimental Design and Evaluation 21
5.1 Experimental Goals 21
5.2 Experimental Setup 21
5.3 Experimental Results: Flood Depth 21
5.4 Experimental Results: Damage Functions 21
5.5 Analysis of Impact on Total Exposed Property Value 21

6 Conclusions & Discussion 23
6.1 Conclusion & Interpretation of Results 23
6.2 Limitations of This Work 23
6.3 Future Work 23

Appendix A Proofs of Theorems 25
A.1 Proof of the Pythagorean Theorem 25

References 27

Bibliography 29

Biography 31

Index 33

List of Figures

Figure 3.1: Example hazard map and depth–return period curve. 8
Figure 3.2: Depth–damage function for residential buildings. 10
Figure 3.3: Expected Annual Damage from flood risk. 12

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem. 25

List of Tables

Table 3.1: Comparison of flood hazard models used in the study. 9

List of Algorithms

Abbreviations

JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School
AEP ... Average Exceedance Probability
EAD ... Expected Annual Damage
ESG ... Environmental, Social, and Governance
JRC ... Joint Research Centre
RP ... Return Period
WRI ... World Resources Institute

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Background and Motivation

Flooding is one of the most frequent and damaging natural hazards globally, responsible for extensive economic losses, social disruption, and loss of life. Climate change is expected to further intensify these risks by increasing both the frequency and severity of hydrological extremes. The Intergovernmental Panel on Climate Change (IPCC) projects a growing incidence of intense precipitation events and flood-related disasters, particularly in regions affected by higher hydrological variability and sea-level rise [1].

In Europe, recent catastrophic flood events—such as those in Germany and Belgium in 2021, and in Slovenia in 2023—have exposed significant vulnerabilities in infrastructure, spatial planning, and emergency preparedness. According to the European Environment Agency (EEA), riverine and pluvial floods already account for more than one-third of total weather-related losses in Europe, a share that is projected to increase over coming decades without improved mitigation and adaptation measures [2].

These developments underscore the urgent need for robust and spatially detailed flood risk models that can support risk-sensitive decision-making in domains such as insurance pricing, urban planning, asset valuation, and regulatory compliance. This is particularly critical for stakeholders in the real estate and financial sectors, who are increasingly required to assess and disclose exposure to climate-related physical risks as part of their environmental, social, and governance (ESG) reporting obligations [3], [4].

1.2 Climate Risk and ESG Disclosure

Climate risk refers to the potential adverse effects of climate change on economic systems, institutions, and physical assets. It is commonly classified into three interrelated categories: *physical*, *transition*, and *liability* risks [5].

- **Physical risks** arise from the direct consequences of climate variability and long-term climate change. These include acute hazards such as floods, storms, and wildfires, as well as chronic effects such as sea-level rise and shifting temperature trends. These risks can cause immediate physical damage and long-term degradation of asset value.

- **Transition risks** arise from the transition to a low-carbon economy. They include policy changes (e.g., carbon pricing, emissions regulations), technology disruption, evolving market preferences, and reputational risks associated with inaction or misalignment.

- **Liability risks** are considered by some [6] and encapsulate risks arising when those affected by anthoropogenic climate change seek compensation.

These categories are increasingly embedded in ESG-aligned regulatory frameworks such as those developed by the Task Force on Climate-related Financial Disclosures (TCFD) and the Network for Greening the Financial System (NGFS). In the European context, the European Central Bank (ECB) requires significant institutions to assess, manage, and disclose climate-related and environmental risks, including physical risks. According to the ECB's supervisory guidance, banks are expected to integrate these risks into their internal governance, risk management, stress testing, and disclosure practices [3].

Physical risk is typically modeled as the interaction of three fundamental components:

- **Hazard** – the probability and intensity of a climate-related event (e.g., flood depth for a 100-year return period),
- **Vulnerability** - the expected damage to an asset given the intensity of the hazard,
- **Exposure** - the presence and value of assets located in potentially affected areas.

These components form the foundation of physical climate risk models used in both regulatory compliance and voluntary disclosures. However, despite the wide adoption of this conceptual framework, practical implementation often suffers from limited model transparency, inconsistent spatial resolution, and poor validation—especially at the local scale relevant to property-level investment and risk management.

This thesis focuses specifically on **acute physical risks**, with a particular emphasis on riverine flooding and its financial impact on real estate assets. By comparing the performance of global and local flood models and evaluating standardized vulnerability functions against observed data, the work aims to contribute to more accurate and transparent risk quantification in support of ESG-aligned decision-making.

1.3 Problem Statement

While global flood hazard models such as the WRI Aqueduct River Floods provide broad coverage and climate scenario integration, they are often limited in spatial resolution and may not reflect local hydrological conditions. On the other hand, high-resolution local models, such as Slovenia's *Integralna karta globin* (IKG), offer detailed inundation mapping but are typically fragmented, static (i.e., not climate scenario-aware), and difficult to integrate into broader risk platforms.

Furthermore, flood vulnerability is typically modeled using generalized depth–damage functions derived from empirical or synthetic data. These functions may perform well in aggregate, but their applicability across different building types, geographies, and damage-reporting standards remains uncertain. This creates uncertainty in downstream financial estimates and limits the usefulness of such models for asset-level ESG disclosure.

1.4 Research Objectives

The purpose of this thesis is to evaluate how well existing flood models can represent real-world flood depths and damage outcomes in Slovenia. Specifically, it compares a global hazard model (WRI Aqueduct) and a high-resolution local model (IKG) to assess the accuracy of their flood depth predictions.

Separately, the thesis tests global depth-damage functions, used to estimate flood-related property value loss, against reported damage data, with the goal of evaluating and potentially improving their performance. Together, these analyses aim to assess the reliability of current modeling tools for understanding physical climate risk and supporting property related decision making in the context of increasing flood exposure.

The main goals of the research are:

1. To improve existing physrisk risk assessment framework with the following contributions:
   (a) Develop a data processing pipeline for creating a unified and homogenous SI IKG flood depth model for different areas of importance.
   (b) Develop a data onboarding pipeline for integrating this new unified model into physrisk framework.
2. To compare the performance of a global flood hazard model (WRI Aqueduct) and a local Slovenian model (IKG) using real flood data.
3. To evaluate how model resolution and proximity to water bodies affect prediction accuracy.
4. To assess the accuracy of global flood depth–damage functions on reported damage cases in Slovenia and improve their performance.

1.5 Structure of the Thesis

The remainder of the thesis is structured as follows:

- **Chapter 2** presents a review of relevant literature and related work on flood hazard modeling, vulnerability estimation, and ESG-aligned risk frameworks.
- **Chapter 3** describes the study area, datasets, and the methodological framework used for model evaluation and data processing.
- **Chapter 4** details the integration of the local IKG model into the Physrisk framework and presents the architecture of the comparison toolset.
- **Chapter 5** provides experimental results, including quantitative evaluation of model performance, depth–damage function analysis, and financial impact estimates.
- **Chapter 6** discusses the key findings, limitations, and possible future research directions, and concludes the thesis.

Chapter 2

Literature Review & Related Work

TBD

Chapter 3

Study Area, Data, and Methodology

This chapter introduces the data sources, mathematical models, and spatial context used to evaluate flood hazard and damage modeling in this thesis. The flood risk assessment is structured around three components: flood hazard (depth), vulnerability (damage functions), and exposure (property value). These components are defined both conceptually and mathematically, and are later integrated into a unified framework for risk estimation and validation.

3.1 Hazard and Depth

Flood hazard refers to the spatial distribution and intensity of flooding under different return periods. In this study, hazard is represented by flood depth, modeled as a spatial function for given exceedance scenarios.

3.1.1 Mathematical Definition of Flood Hazard

Flood hazard is modeled as a spatially varying flood depth field:

$$h : \Omega \times \mathcal{T} \to \mathbb{R}_{\geq 0}, \quad (x, y, T) \mapsto h(x, y, T) \tag{3.1}$$

where:

- $\Omega \subset \mathbb{R}^2$ is the spatial domain of interest,
- $\mathcal{T} = \{T_1, T_2, \ldots, T_K\}$ is the set of return periods considered (e.g., 10, 100, 500 years),
- $h(x, y, T)$ is the flood depth (in meters) at location (x, y) for return period T.

Return periods T express the average time between events of a given magnitude. The associated annual exceedance probability p is:

$$p = \frac{1}{T} \tag{3.2}$$

Flood hazard models estimate water depths $h(x, y, T)$ using hydrological and hydraulic simulations that account for catchment runoff, river dynamics, topography, and in some cases, climate change. In practice, these functions are discretized into raster data aligned to a grid. The continuous function $h(x, y, T)$ is approximated by an array $H_{i,j,k}$, where:

- i, j index the spatial grid cells,
- k indexes the return period $T_k \in \mathcal{T}$.

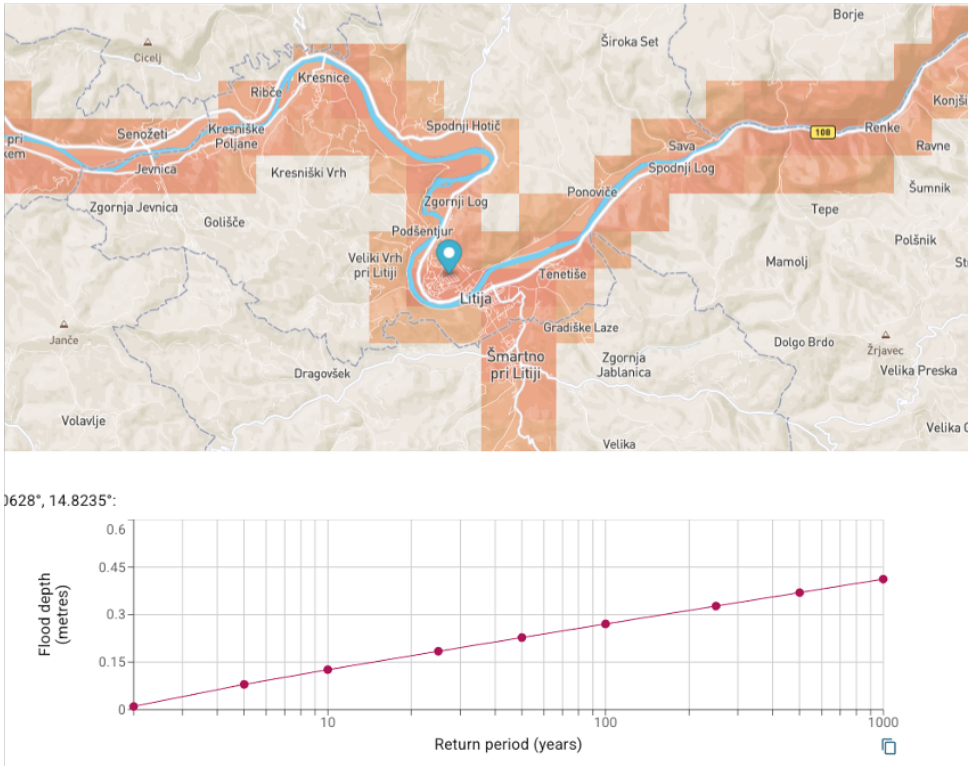

Figure 3.1: Flood hazard illustration for a location in Slovenia (Litija). The top panel shows the spatial extent of riverine flood modeling; the bottom panel plots flood depth $h(x, y, T)$ as a function of return period T, demonstrating the monotonic growth of hazard with increasing return period.

3.1.2 Hazard Data Sources and Modeling Approach

Two flood hazard models are evaluated and compared in this thesis:

- **WRI Aqueduct River Flood Model:** This is a global flood hazard dataset developed by the World Resources Institute. It estimates flood depths using hydrological and hydraulic simulations from global river basins, based on historical river discharge, elevation models (e.g., MERIT DEM), and land surface parameters. The model provides depth estimates for return periods ranging from 2 to 1000 years, with global coverage at 1 km resolution. Critically, it supports multiple future climate scenarios via Representative Concentration Pathways (RCPs), specifically RCP4.5 and RCP8.5. These scenarios reflect moderate and high greenhouse gas emissions trajectories, respectively:

  - **RCP4.5** assumes stabilization of radiative forcing at 4.5 W/m^2 by 2100, representing moderate mitigation.
  - **RCP8.5** assumes continued high emissions, leading to >4°C global warming by 2100.

  Climate scenarios affect rainfall intensities, runoff, and river discharge, which in turn affect simulated flood extents and depths. The outputs are published in GeoTIFF raster format, using EPSG:4326 (WGS84).

- **Integralna karta globin (IKG):** This high-resolution local hazard dataset is developed by Slovenian civil protection authorities and technical consultants. It includes flood hazard maps for selected municipalities based on detailed hydraulic modeling (1D/2D simulations) calibrated using historical events. Flood extents are modeled under synthetic return period scenarios (10, 100, 500 years) and expressed in three discrete flood depth classes:

  - GM – minor inundation (up to 0.5 m),
  - GS – significant inundation (up to 1.5 m),
  - GB – severe inundation (up to 2.5 m).

  IKG data are originally stored as vector polygons (shapefiles or GeoPackages) using the national coordinate system (EPSG:3794). For integration and analysis, they are rasterized to a 5 m grid, harmonized with other layers, and stored in Zarr format. Unlike WRI, IKG models do not include climate scenario adjustments.

Table 3.1: Comparison of flood hazard models used in the study.

Feature	**WRI Aqueduct River Flood**	**Integralna karta globin (IKG)**
Spatial resolution	∼1 km	5 m
Coverage	Global	18 regions in Slovenia
Climate scenarios	RCP-aware (RCP4.5, RCP8.5)	Not scenario-aware
Depth type	Continuous (in meters)	Discrete classes (0.5, 1.5, 2.5 m)
Return periods	2 to 1000 years	10, 100, 500 years
File format	Raster (GeoTIFF)	Vector (shapefile, GPKG)
Coordinate system	EPSG:4326	EPSG:3794 (transformed)

Both datasets were transformed to a common 5 m grid and reference system to enable cell-wise comparisons of depth predictions and derived risk estimates.

3.2 Vulnerability and Damage Function

Flood vulnerability quantifies the expected damage to an asset given the intensity of the flood at that location. This relationship is typically modeled via a depth–damage function.

3.2.1 Mathematical Definition of Flood Vulnerability

A depth–damage function $D(h)$ maps flood depth h to the expected fractional loss in asset value:

$$D : \mathbb{R}_{\geq 0} \to [0, 1], \quad h \mapsto D(h) \tag{3.3}$$

The function $D(h)$ is assumed to be monotonic and piecewise continuous. It starts at a nonzero depth threshold (typically 0.1–0.2 m), increases with water depth, and saturates at a maximum value (e.g., 0.7 to 0.9) depending on asset type.

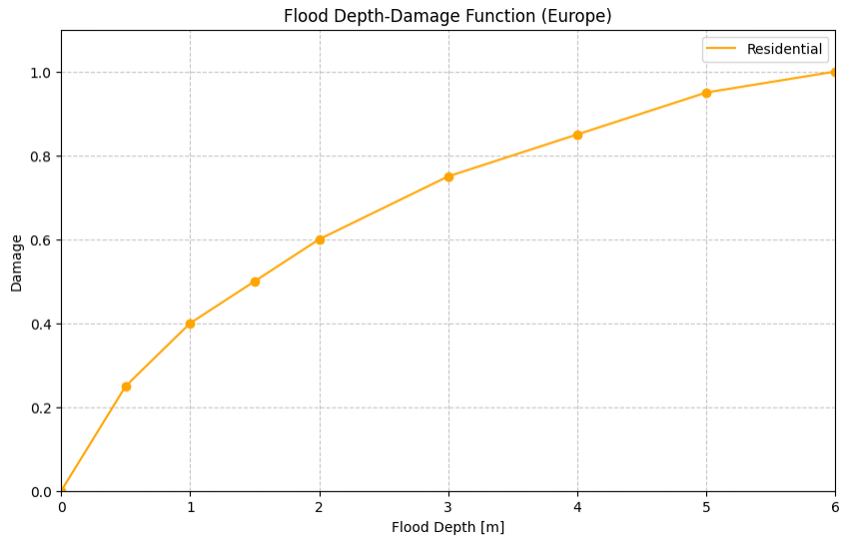

Figure 3.2: Standardized depth–damage function for residential buildings in Europe, as published by the Joint Research Centre (Huizinga et al., 2017). The curve shows how expected damage increases with flood depth h, up to a saturation point near total loss.

3.2.2 Vulnerability Data Source

This thesis uses standardized global depth–damage functions published by Huizinga et al. [7], developed under the auspices of the European Commission Joint Research Centre (JRC). These functions are stratified by building type (e.g., residential, commercial, industrial) and were applied based on available metadata in the Slovenian dataset.

Damage reports from the Administration for Civil Protection and Disaster Relief of Slovenia serve as the primary reference for validation. These records include georeferenced flood locations, reported water depths, and economic losses. When available, building classifications were matched to appropriate JRC vulnerability curves.

3.3 Flood Risk Modeling

Flood risk refers to the expected financial loss due to flooding and is commonly expressed as a function of three interacting components: hazard, vulnerability, and exposure. This section presents a mathematical framework for calculating risk at the asset level and aggregating it across space and probability. It distinguishes between loss for a specific return period and Expected Annual Damage (EAD), which aggregates over all relevant scenarios.

3.3.1 Risk for a Single Return Period

For a given return period T, the expected loss at a spatial location (x, y) is defined as:

$$R(x, y; T) = D(h(x, y, T)) \cdot E(x, y) \tag{3.4}$$

where:

- $h(x, y, T)$ is the modeled flood depth (hazard),
- $D(h)$ is the depth–damage function (vulnerability),
- $E(x, y)$ is the value of the asset at location (x, y) (exposure).

This formulation reflects a scenario-based approach. For example, consider a property valued at €100,000. If the hazard model estimates a flood depth of $h = 0.31\,\text{m}$ for the 100-year return period ($T = 100$), and the damage function gives $D(0.31) = 0.17$, then the expected loss is:

$$R(x, y; 100) = 0.17 \cdot 100{,}000\,€ = 17{,}000\,€$$

This estimate represents the expected damage conditional on a specific flood event.

To obtain total loss across a spatial region Ω, the losses are integrated (or summed for raster data):

$$R_{\text{total}}(T) = \int_{\Omega} D(h(x, y, T)) \cdot E(x, y)\, dx\, dy \tag{3.5}$$

$$R_{\text{total}}(T_k) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} D(H_{i,j,k}) \cdot E_{i,j} \tag{3.6}$$

This discretized version is applied in this thesis to compare risk under different hazard models (WRI vs IKG) at specific return periods.

3.3.2 Expected Annual Damage (EAD)

While scenario-based losses (e.g., for RP100) are useful, they do not fully describe the stochastic nature of flood risk. A more comprehensive measure is the **Expected Annual Damage (EAD)**, which integrates risk across the full distribution of return periods or annual exceedance probabilities (AEPs).

Let $p \in (0, 1]$ denote the AEP corresponding to return period $T = 1/p$. Then the EAD at location (x, y) is given by:

$$\text{EAD}(x, y) = \int_0^1 D(h(p)) \cdot E(x, y)\, dp \tag{3.7}$$

In practice, this is approximated using trapezoidal integration over a discrete set of modeled return periods:

$$\text{EAD}(x, y) \approx \sum_{k=1}^{K-1} \frac{p_k - p_{k+1}}{2} \cdot [R(x, y; T_k) + R(x, y; T_{k+1})]$$

where $\{T_k\}$ are the available return periods and $p_k = 1/T_k$. This method captures both frequent and rare events in a single, annualized indicator.

3.3.3 Exposure Data

Exposure refers to the monetary value of physical assets potentially affected by flooding. For the Slovenian case study, the exposure values were derived from:

- Appraised damage reports submitted to civil protection authorities,
- Official cadastral or tax-assessed values (when available),
- Synthetic estimates based on building typology and floor area.

All exposure estimates were spatially harmonized with hazard and vulnerability layers and aligned to a common grid. Values were normalized where necessary to match the assumptions of the damage function calibration (e.g., replacement cost basis).

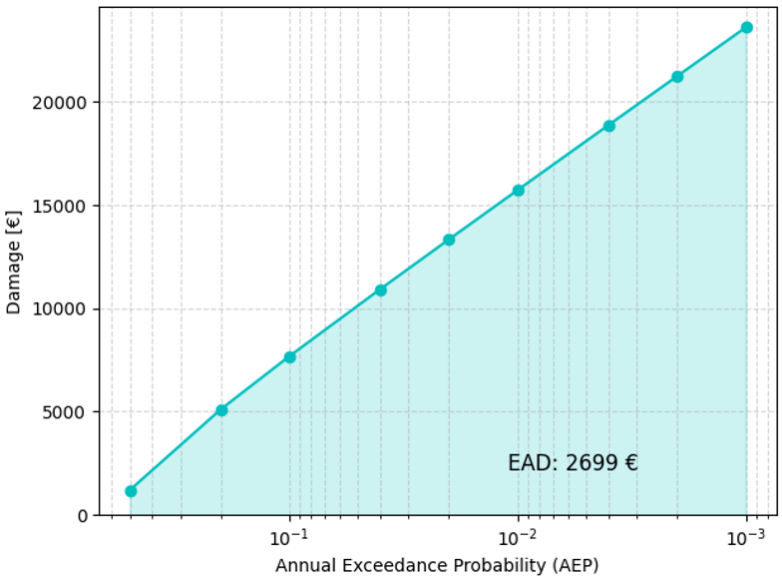

Figure 3.3: Illustration of Expected Annual Damage (EAD) calculation. The curve plots flood damage against annual exceedance probability; the shaded area under the curve represents the EAD for a given asset.

3.3.4 Use Cases

This risk modeling framework supports multiple analytical and practical applications:

- **Model comparison:** Evaluate spatial accuracy of WRI vs IKG hazard inputs.
- **Vulnerability validation:** Assess the applicability of global depth–damage functions using local empirical data.
- **EAD calculation:** Derive annualized loss estimates for integration into financial and regulatory workflows.
- **Sensitivity analysis:** Examine the influence of spatial resolution and waterbody proximity on model performance.
- **Platform integration:** Enable modular onboarding into the Physrisk API and ESG disclosure platforms.

Chapter 4

Model Integration and Analysis Framework

Accurate flood risk assessment requires the integration of diverse data sources and computational tools to simulate hazard intensity, estimate potential damages, and evaluate financial exposure. In this chapter, we present the technical framework developed to combine heterogeneous flood hazard data—particularly local Slovenian models—into a unified system suitable for analysis and comparison within the Physrisk platform.

The focus lies on harmonizing the *Integralna karta globin* (IKG) flood maps, which vary in structure, format, and coordinate reference systems, and transforming them into a consistent, machine-readable representation. These models are rasterized, standardized by return period, and encoded in the cloud-optimized Zarr format. The processed datasets are then onboarded into the modular Physrisk platform, which separates hazard, vulnerability, and financial components and allows for flexible model evaluation.

To facilitate empirical comparison, the framework also includes a toolset that enables side-by-side querying and batch evaluation of both global and local models at precise geolocations, aligned with the validation dataset. This infrastructure supports all subsequent experimental analysis presented in Chapter 5 and lays the groundwork for scalable, reproducible climate risk assessments.

The chapter is organized as follows:

- Section 4.1 presents the overall system architecture and the interaction between components.
- Section 4.2 describes the pipeline for transforming and combining IKG models across multiple Slovenian regions.
- Section 4.3 explains how the standardized hazard data are integrated into the Physrisk platform.
- Section 4.4 outlines the model comparison toolset used to evaluate prediction accuracy across models.

4.1 Architectural Diagram

The integration of global and local flood hazard models into a unified analysis platform requires a modular, extensible architecture that supports heterogeneous input formats, scenario-aware modeling, and scalable evaluation. This section presents the architecture developed for this purpose, highlighting its components, data flow, and interoperability within the *Physrisk* framework.

4.1.1 System Overview

Figure **??** illustrates the architecture of the flood hazard model integration pipeline, based on the open-source *OS-Climate Physrisk* ecosystem. The process begins with raw flood hazard data, which may originate from global sources such as the WRI Aqueduct or TU Delft flood models, or from local sources such as the Integralna karta globin (IKG) used in Slovenia. These datasets are onboarded into the `os-risk-hazard` module, a Python-based processing library designed to standardize, harmonize, and prepare hazard indicators for risk evaluation.

The onboarded models are then encoded in Zarr format and stored in an AWS S3-compatible object storage. The Physrisk framework retrieves hazard indicators from this backend and combines them with vulnerability and exposure components to compute impact distributions. The resulting outputs can be accessed via the Physrisk API or used directly for ESG reporting and financial risk analysis through the `os-risk-financial` module.

4.1.2 Data Flow and Model Interoperability

Each flood model—whether global or local—is processed through the same onboarding interface, allowing standardization of outputs while preserving return-period-specific intensity information. The hazard models are structured into three-dimensional Zarr arrays indexed by latitude, longitude, and return period. This enables consistent querying and evaluation across models with different spatial resolutions or geographic coverage.

Once onboarded, all hazard layers become interchangeable within the Physrisk pipeline. Vulnerability models (e.g., depth–damage functions) and exposure datasets (e.g., building values or asset locations) can then be used in combination with any hazard source. This interoperability facilitates both comparative experiments and compound risk evaluations, where multiple models are used to bound uncertainty or stress-test assumptions.

4.1.3 Integration of the IKG Model

The locally developed IKG model, originally provided in shapefile and GeoPackage vector formats, required a dedicated pre-processing stage to align with the Physrisk architecture. This process involved:

- Transforming coordinate systems (from EPSG:3794 to EPSG:4326),
- Rasterizing vector layers into a fixed-resolution 5x5m grid,
- Assigning discrete flood depth values to each return period (RP10, RP100, RP500),
- Converting the raster layers into Zarr format for chunked cloud storage.

The unified model was then uploaded to AWS S3 and integrated into the Physrisk hazard module alongside WRI and TU Delft models. This enabled direct comparison and batch evaluation of global versus local model predictions, described further in Section 4.4.

4.1.4 Hazard Onboarding Pipeline

The transformation from raw IKG data into a cloud-optimized flood hazard model is summarized in Figure **??**. The pipeline consists of the following steps:

1. **Rasterization:** Converting input vector data into gridded raster tiles using a 5x5m resolution grid.

2. **Zarrify:** Chunking the raster into 1000x1000 tiles and encoding into Zarr format.
3. **Upload:** Transferring the data to AWS S3 for use within Physrisk.

The final onboarded dataset can be queried via the Physrisk API or accessed directly for integration with ESG tools or batch model evaluation.

4.2 Pipeline for Combining Heterogeneous IKG Models

The *Integralna karta globin* (IKG) flood hazard datasets are developed for individual municipalities or regions in Slovenia, each based on local hydrological and hydraulic analyses. These datasets differ in file structure, geospatial reference systems, data quality, and encoding of flood depths. To integrate these into a unified flood hazard model suitable for comparative evaluation and platform onboarding, a standardized processing pipeline was developed.

4.2.1 Data Sources and Heterogeneity

The raw IKG datasets were obtained from multiple engineering consultancies (*Inženiring za vode d.o.o., Urbania d.o.o., iS Projekt d.o.o., IZVOR d.o.o., DHD d.o.o.*), each responsible for flood modeling in a subset of areas classified as "Območja pomembnega vpliva" (Areas of Significant Flood Risk) by Slovenian water authorities.

Each local model typically includes:

- GIS vector data (shapefiles, GeoPackages) representing inundation zones,
- Attribute data encoding flood extent and classes for selected return periods (e.g., 10-, 100-, and 500-year events),
- Region-specific coordinate systems (typically EPSG:3794, Slovenia National Grid).

In total, data from 18 areas of importance were collected, with each area providing flood scenarios for three return periods. This resulted in 54 distinct layers that needed to be harmonized.

4.2.2 Standardization and Processing Workflow

To unify the IKG datasets into a single model, the following pipeline was developed and implemented in Python using open-source geospatial processing libraries (GDAL, Fiona, Rasterio, xarray):

1. **Coordinate Transformation:** All source files were transformed from EPSG:3794 to WGS84 (EPSG:4326) to ensure compatibility with global models and standardized grid alignment.
2. **Rasterization:** Vector layers were rasterized into a regular grid with a resolution of 5 meters (5x5m), which approximates the native resolution of the original hydrological models. A fixed flood depth value was assigned to each flood class:
    - GM (minor flooding): 0.5 m
    - GS (significant flooding): 1.5 m
    - GB (severe flooding): 2.5 m

3. **Return Period Alignment:** Each rasterized layer was tagged with its associated return period (RP10, RP100, RP500). For each area, three separate rasters were produced and stacked by return period.

4. **Tiling and Merging:** The processed rasters from all areas were mosaicked into a common spatial grid and stored as a 3D array indexed by latitude, longitude, and return period. Conflicts in overlapping regions (e.g., model boundaries) were resolved by assigning priority to higher-confidence datasets based on metadata quality.

4.2.3 Unified Flood Hazard Product

The final result of this pipeline is a unified flood hazard model for Slovenia covering all 18 areas of importance. This model:

- Captures flood depth estimates at high spatial resolution for three return periods,
- Standardizes input data from diverse local models into a single harmonized format,
- Enables consistent comparison with global models such as WRI Aqueduct within the same framework.

Figure **??** provides a schematic representation of the transformation process, from raw shapefiles to a standardized hazard product.

This unified model was subsequently converted to Zarr format and onboarded into the Physrisk platform, as described in Section 4.3.

4.3 Physrisk Framework – Model Onboarding

Once the local IKG flood models were unified into a standardized and spatially consistent format, the next step involved integrating them into the modular *Physrisk* framework. Physrisk is part of the broader OS-Climate open-source ecosystem and is designed to facilitate scalable, scenario-based physical climate risk analysis. The framework separates hazard, vulnerability, and financial components and enables flexible model composition, comparison, and evaluation.

4.3.1 Physrisk Architecture and Modules

The Physrisk architecture is composed of several interoperable modules:

- **OS-risk-hazard**: responsible for ingesting and formatting hazard data sets (e.g., flood depth by return period) into a common format.
- **OS-risk-physrisk**: computes impact distributions by convolving hazard data with vulnerability functions and exposure information.
- **OS-risk-financial**: translates physical impacts into financial metrics such as loss of asset value or disruption to operations.
- **Physrisk API**: exposes model outputs through a standardized programmatic interface.

The onboarding of hazard models occurs within the `os-risk-hazard` component and requires the hazard data to conform to a Zarr-based storage structure optimized for parallel access and spatial querying.

4.3.2 Zarr Format and Indexing

The Zarr format is a cloud-native format for storing large multi-dimensional arrays. It was chosen for hazard model storage due to its ability to:

- Efficiently compress and store gridded data in chunks (e.g., 1000×1000 tiles),
- Allow selective download of spatial subsets,
- Support multi-indexing (e.g., latitude, longitude, return period).

For the unified IKG model, the hazard data were encoded into a 3-dimensional Zarr array with the following dimensions:

- i: Latitude (y-axis),
- j: Longitude (x-axis),
- k: Return period index, corresponding to RP10, RP100, and RP500.

Each tile was stored as a compressed chunk and uploaded to an AWS S3-compatible bucket accessible to all Physrisk modules.

4.3.3 Onboarding Procedure

The onboarding process was implemented using the Physrisk hazard onboarding tool, a Python-based utility designed for structured hazard model integration. The steps were as follows:

1. **Prepare Metadata:** Define the return periods, CRS (EPSG:4326), spatial resolution, and transformation matrix for the IKG model.
2. **Convert to Zarr:** Use the onboarding tool to encode the processed GeoTIFF files into a Zarr store, assigning appropriate chunk sizes (e.g., 1000×1000 pixels).
3. **Upload to Cloud Storage:** Transfer the Zarr directory to a cloud-accessible S3 bucket used by the OS-Climate infrastructure.
4. **Register the Model:** Add a model reference entry into the hazard configuration of Physrisk, enabling downstream access via API and batch processors.

4.3.4 Integration with Other Models

With the IKG model onboarded, it became available in the same infrastructure alongside global flood hazard layers, including:

- WRI Aqueduct River Flood (global, 1 km resolution),
- TU Delft River Flood (Pan-European, 100 m resolution).

This interoperability allows any of the onboarded hazard layers to be used with common vulnerability functions and exposure datasets, enabling direct comparisons, uncertainty bounding, and ensemble evaluations. The IKG model is now fully accessible via the Physrisk API and can be used for property-level flood risk analysis in Slovenia under a unified modeling interface.

4.4 Comparison Toolset

To evaluate and compare the predictive performance of flood hazard models, a model-agnostic comparison toolset was developed and integrated into the Physrisk framework. This toolset enables side-by-side querying of multiple hazard sources at specific geolocations and return periods, aligned with the reference flood observations used for validation in Chapter 5.

4.4.1 Purpose and Design Principles

The comparison toolset was designed with the following goals in mind:

- **Model-agnostic access:** Allow querying of different onboarded hazard models (e.g., IKG, WRI Aqueduct) through a unified interface.
- **Return-period specificity:** Ensure comparability by extracting flood depth predictions at the same return period (e.g., 100-year flood).
- **Location-accurate evaluation:** Retrieve model outputs at high spatial precision to match reported flood locations.
- **Batch processing support:** Enable evaluation across hundreds of observations using efficient geospatial access patterns.

4.4.2 Interface and Functionality

The toolset consists of a lightweight Python interface built on top of the Physrisk hazard API. Given a set of coordinates (latitude, longitude) and a desired return period, the tool performs the following steps:

1. **Coordinate Transformation:** Convert input coordinates into the grid index of the onboarded Zarr array using affine transformations defined in the model metadata.
2. **Chunk Identification and Retrieval:** Locate the Zarr chunk containing the requested point and load the corresponding depth value for the specified return period.
3. **Output Formatting:** Return the flood depth value, return period, and model metadata for downstream use in regression or classification analysis.

When applied in batch mode, the tool leverages parallel reads and lazy loading to evaluate large datasets (e.g., the full validation set of flood reports) with minimal memory overhead.

4.4.3 Supported Models and Inputs

The toolset was validated for use with the following onboarded hazard models:

- **SI IKG (Unified Model)**: High-resolution local flood model rasterized at 5x5 m, covering 18 Slovenian areas of importance, with RP10, RP100, and RP500 layers.
- **WRI Aqueduct Flood Hazard**: Global model with 1 km resolution, scenario-aware, supporting multiple return periods from RP2 to RP1000.

The input to the toolset is a tabular dataset containing:

- Latitude and longitude of each observation,
- Desired return period (e.g., RP100),
- Optional fields: timestamp, municipality, reported depth, reported damage.

4.4.4 Integration with Validation Workflow

The comparison outputs are stored in a structured table, allowing for direct computation of model performance metrics such as:

- **Regression error:** difference between predicted and observed flood depths,
- **Classification accuracy:** match rate within flood depth categories (e.g., <0.5 m, 0.5–1.5 m, >1.5 m),
- **Bias analysis:** mean and median error segmented by proximity to water, terrain type, or municipality.

These outputs form the quantitative basis for the experimental results presented in Chapter 5.

4.4.5 Reproducibility and Extensibility

The toolset was developed as a standalone module and version-controlled as part of the project's repository. This design ensures that future updates to hazard models (e.g., inclusion of climate scenario variants or additional regions) can be incorporated without changes to the validation logic.

Furthermore, the modularity of the toolset supports extension to:

- Additional hazard types (e.g., pluvial flooding),
- Vulnerability assessments (via model chaining),
- Scenario-based risk projections (by switching hazard layers).

Thus, the comparison toolset serves not only as a core utility for this thesis, but also as a scalable evaluation framework for broader applications in physical climate risk modeling.

Chapter 5

Experimental Design and Evaluation

TBD

5.1 Experimental Goals

TBD

5.2 Experimental Setup

TBD

5.3 Experimental Results: Flood Depth

TBD

5.4 Experimental Results: Damage Functions

TBD

5.5 Analysis of Impact on Total Exposed Property Value

TBD

Chapter 6

Conclusions & Discussion

6.1 Conclusion & Interpretation of Results

TBD

6.2 Limitations of This Work

TBD

6.3 Future Work

TBD

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page **??**.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

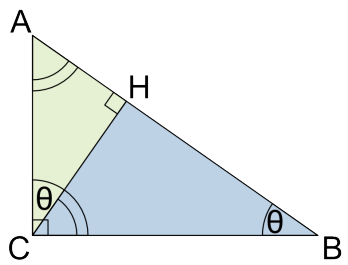

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

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Publications Related to the Thesis

All publications related to the thesis should be referenced in the text.

Other Publications (optional)

...

Biography

The author of this thesis ...

Index

Pythagorean theorem
    proof, 25